Wenqian Xing

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Research Interests

My research interests broadly lie in stochastic modeling and decision-making under uncertainty, with a focus on causal inference, reinforcement learning, and AI evaluation and alignment.

Education

Stanford University 2023 – 2027 (Expected)
Ph.D. in Operations Research
M.S. in Management Science and Engineering

Columbia University 2021 – 2023
M.S. in Operations Research

University of Michigan – Shanghai Jiao Tong University 2016 – 2021
M.S. in Financial Engineering
B.S. in Electrical and Computer Engineering; B.S. minor in Data Science

Preprints

(α-β): authors in alphabetical order

[1] Augmented Truncated Policy Gradients for Rapid-Switching Experiments
W. McCloskey, W. Xing, A. Badanidiyuru, E. Becker, C. Harris, S. Wager, R. Johari
CODE@MIT 2026, Deployed at Uber

[2] Fast Access or Familiar Care? Learning-Based Capacity Allocation with Endogenous Patient Choice
W. Xing, Y. Zhang, Y. Hu
Under review at *Management Science*

[3] Learning to Communicate Under Alignment Uncertainty: A Bayesian Persuasion Bandit Approach
W. Xing, R. Johari
ML×OR Workshop, NeurIPS 2026
Finalist, 2026 DMDA Workshop Best Paper Competition (Theoretical Track)

[4] Online Learning for Dynamic Service Mode Control
W. Xing, Y. Hu, A. Kalvit, V. Sarhangian
Major revision at *Operations Research*
Finalist, 2026 INFORMS Service Science Best Student Paper Competition

[5] Generalized Hypercube Queueing Models with Overlapping Service Regions
W. Xing, S. Zhu, Y. Xie
Major revision at *Service Science*
First Place, Clifford Spiegelman Student Paper Competition, JSM 2026

[6] Optimizing Server Locations in Spatial Queues: Parametric and Non-parametric Bayesian Optimization Approaches
(α-β) C. Hua, A. J. Swersey, W. Xing, Y. Zhang
Major revision at *Operations Research*

[7] Attention-Limited Reward Learning
W. Xing

Publications

[1] Estimation of Treatment Effects under Nonstationarity via the Truncated Policy Gradient Estimator
(α-β) R. Johari, T. Peng, W. Xing
Under review at *Stochastic Systems*
ICML 2026
ML×OR Workshop, NeurIPS 2025

[2] Conservative Inference in Switchback Experiments
(α-β) J. Blanchet, P. W. Glynn, R. Johari, L. Wu, W. Xing
AISTATS 2026

[3] Black-Box Optimization with Implicit Constraints for Public Policy
W. Xing, J. Lee, C. Liu, S. Zhu
Oral Paper (4.9%), AAAI 2025
Refereed Paper, INFORMS Optimization Society Conference 2024

[4] Multi-Priority Patient Transfer Under Ambulance Offload Delay
W. Xing, C. Hua
Production and Operations Management 2025
Finalist, Best Student Paper Competition, POMS-China International Conference 2024
Second Place, Best Student Paper Competition, INFORMS Conference on Service Science 2021

[5] Optimal Unit Locations in Emergency Service Systems with Bayesian Optimization
W. Xing, C. Hua
Finalist, Best Paper Competition, INFORMS Conference on Service Science 2022
Finalist, Best Student Paper Competition, POMS-HK International Conference 2022

[6] NuSC: An Effective Local Search Algorithm for Solving the Set Covering Problem
C. Luo, W. Xing, S. Cai, C. Hu
IEEE Transactions on Cybernetics 2022

[7] Predictive Job Scheduling under Uncertain Constraints in Cloud Computing
H. Dong, B. Wang, B. Qiao, W. Xing, C. Luo, S. Qin, Q. Lin, D. Zhang, G. Virdi, T. Moscibroda
IJCAI 2021, Deployed at Microsoft Azure

[8] Correlation-Aware Heuristic Search for Intelligent Virtual Machine Provisioning in Cloud Systems
C. Luo, B. Qiao, W. Xing, X. Chen, P. Zhao, C. Du, R. Yao, H. Zhang, W. Wu, S. Cai, B. He, S. Rajmohan, Q. Lin
AAAI 2021, Deployed at Microsoft Azure

Industrial Experiences

Amazon Jun 2026 – Sep 2026
Applied Scientist Intern, Amazon Core AI
– Conducted research on multi-agent decision-making for Amazon Stores

Millennium Management Aug 2021 – Oct 2021
Quantitative Research Intern, WorldQuant
– Sought trading signals using abnormal data in financial markets

Cardinal Operations Jul 2021 – Aug 2021
Algorithm Research Intern
– Applied and tested the COPT mathematical optimization solver

Microsoft Mar 2020 – Jul 2021
Research Intern, Microsoft Research Asia
– Conducted research on online resource allocation for Microsoft Azure
– Developed heuristic solvers for NP-hard problems, e.g., SAT, MAXSAT, and Set Covering

Intel Sep 2019 – Dec 2019
Capstone Project, Intel APAC R&D
– Completed bachelor's thesis, "Deep Learning Foundation in FFmpeg"

Teaching

Stanford: MS&E 226 Fundamentals of Data Science	2025, 2026
Stanford: OIT 245 Optimization and Simulation Modeling	2024
Columbia: IEOR 4106 Stochastic Models; IEOR 4407 Game Theoretical Models	2022
UM-SJTU: VG 441 Optimization Methods in Supply Chain	2020

Awards

Student Scholarship, INFORMS Workshop on Data Science	2022, 2024
MS&E Graduate Fellowship, Stanford University	2023
IEOR Graduate Fellowship, Columbia University	2022
Professional Development Scholarship, Columbia University	2022
Outstanding Graduate Award, Shanghai Ministry of Education	2021
Future Talent Award, Microsoft Research	2020

Service

Committee Observer, George B. Dantzig Dissertation Award
Program Committee Member, INFORMS Workshop on Data Science
Webmaster, Stochastic Networks, Applied Probability, and Performance (SNAPP) Seminar Series
Session Chair, "Experimentation and Interference", INFORMS Annual Meeting 2025
Reviewer, *NeurIPS, ICML, AAAI, AISTATS, IEEE Transactions on Neural Networks and Learning Systems, Production and Operations Management, Amazon Machine Learning Conference*